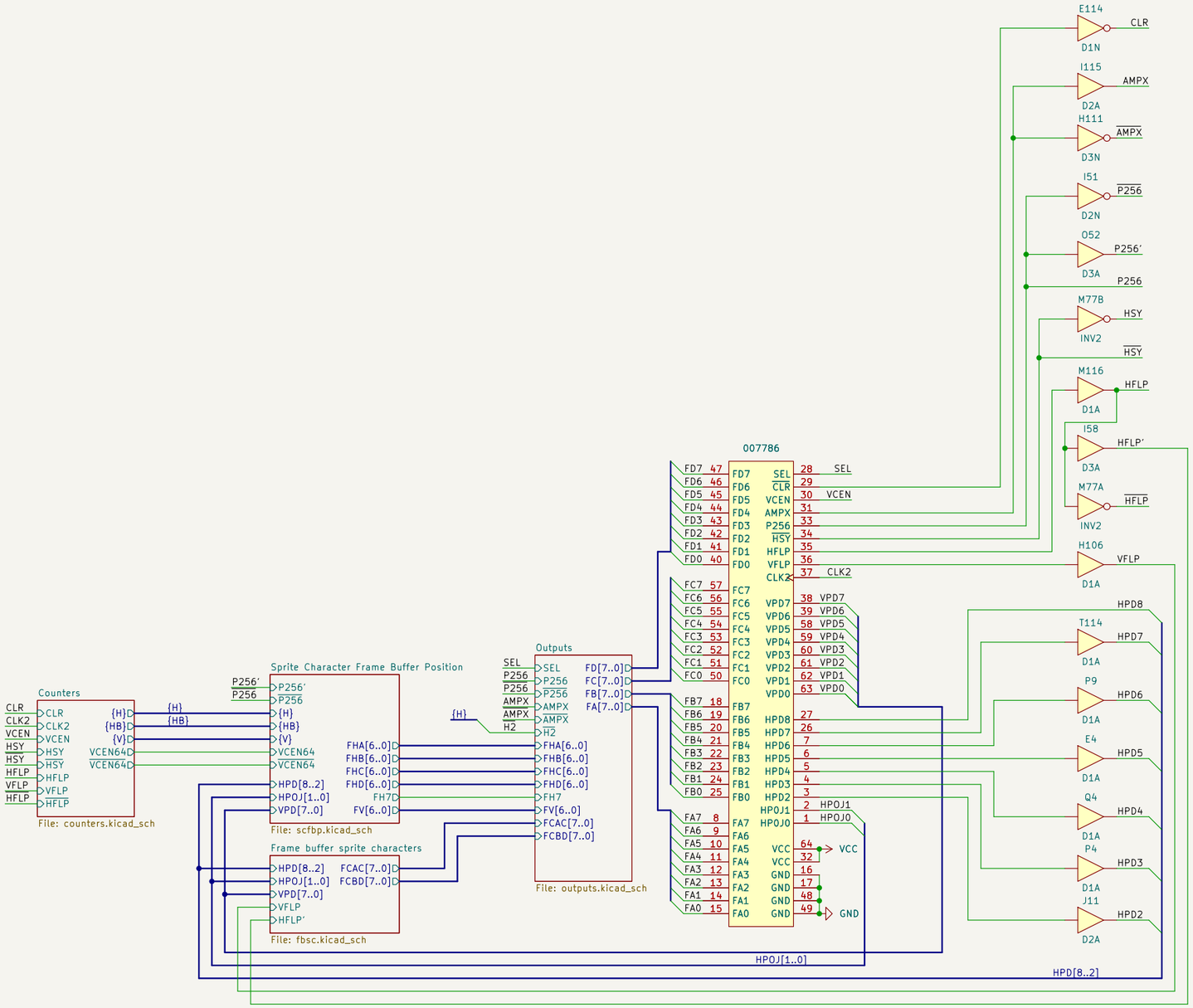


The 007786 is a sprite frame buffer controller.
It works in two modes:
– When SEL = 0, it generates addresses for the frame buffer.
– When SEL = 1, it parses sprite character data
It also handles refreshing of data in the DRAM memory.

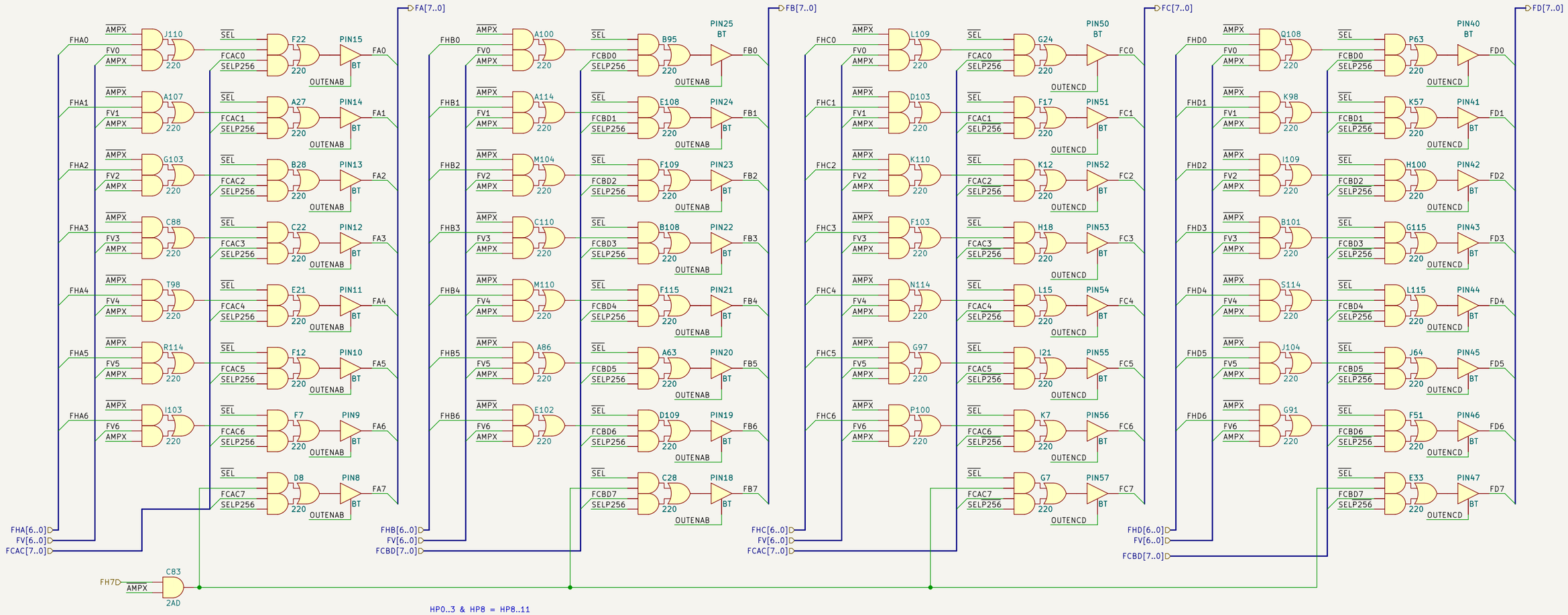
The 007786 uses an OKI 74V000 CMOS gate array
with 2400 unit cells.

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Sheet: /
File: 007786.kicad_sch

Title: Konami 007786

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HP0..3 & HP8 = HP8..11

Output buffer enable

Addressing mode, SEL = 0

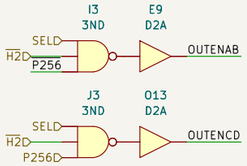
Outputs are enabled all the time when the 007786 is used for addressing sprite characters in the frame buffer, SEL = 0.

Sprite character mode, SEL = 1

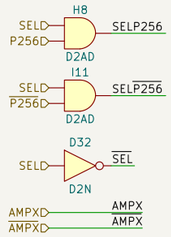
When used for writing sprite characters to the frame buffer it is enabled when H2 = 1 or for FA and FB outputs when P256 = 1. The corresponding for C and D outputs is when P256 = 0.

A and B output
OUTENAB_= ~SEL | ~H2 | ~P256 = ~SEL | H2 | P256

OUTENCD_= ~SEL | ~H2 | ~P256 = ~SEL | H2 | P256



SCM, P256 = 1, AB out enabled.



AMPX and $\overline{AMPX}$ selects between Row and Columns in the frame buffer DRAM.

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Sheet: /Outputs/
File: outputs.kicad_sch

Title: Konami 007786

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V1F is synchronized to $\overline{HSY}$ while the other vertical counters are synchronized to VCEN.

H16B is the same as H16 but shifted one half H8 period later (90 deg).

H8B = ~H8

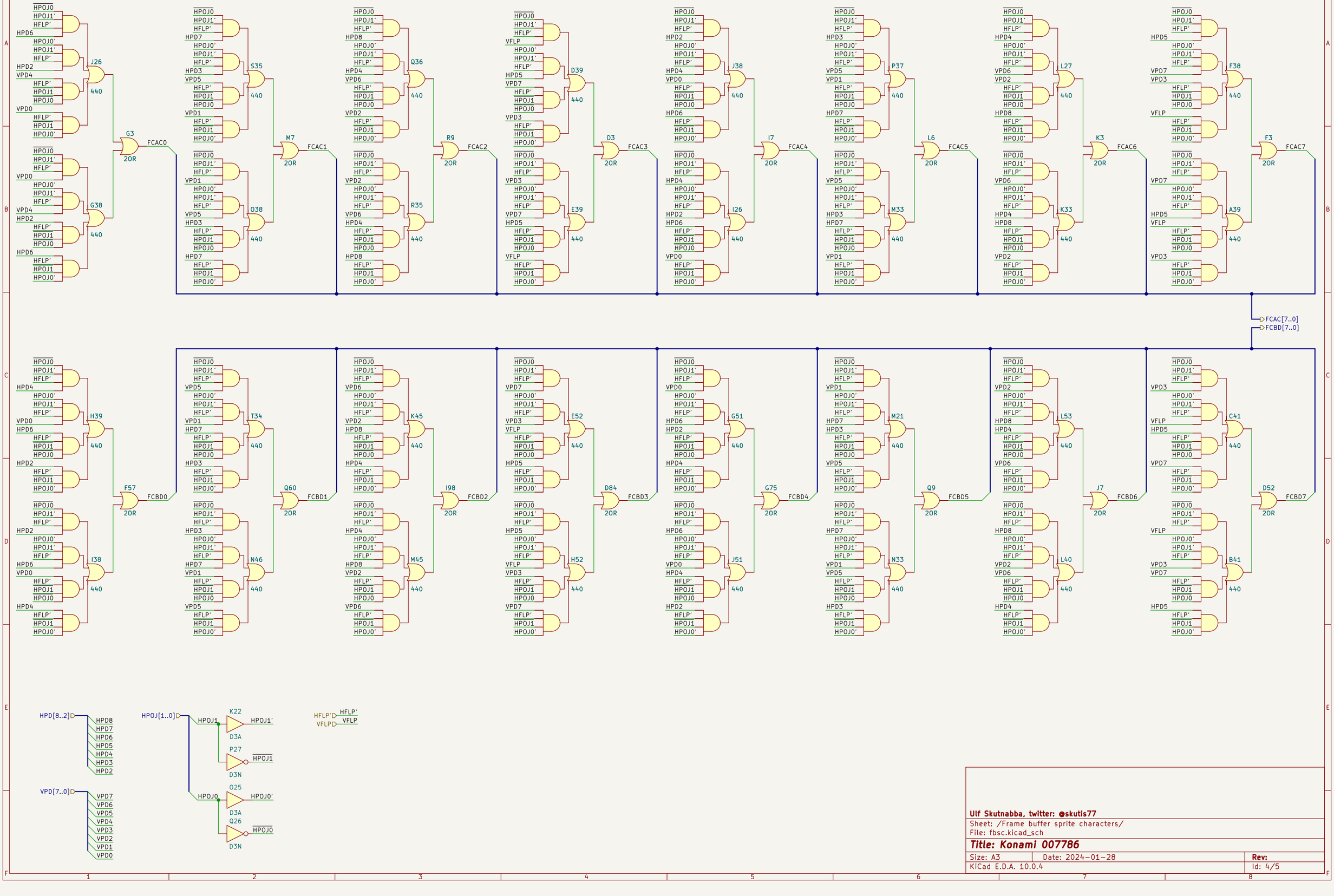
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Sheet: /Counters/

File: counters.kicad_sch

Title: Konami 007786

Size: A3	Date: 2024-01-28	Rev:
KiCad E.D.A. 10.0.4		Id: 3/5



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Sheet: /Frame buffer sprite characters/
File: fbsc.kicad_sch

Title: Konami 007786

Size: A3	Date: 2024-01-28	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/5

Horizontal Position

$\overline{\text{VCEN64}}$ active: Data is written to the frame buffer
VCEN64 active, four times pre frame: Data is refreshed in DRAM, every 4ms.

HP1 to HP4

HP4..7

HP2 to HP5

HP3 to HP6

HP4 to HP7

Vertical Position

P256′ active: Write addressing
P256 active: Read addressing

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Sheet: /Sprite Character Frame Buffer Position/
File: scfbp.kicad_sch

Title: Konami 007786

Size: A3 | Date: 2024-01-28 | Rev:
KiCad E.D.A. 10.0.4 | Id: 5/5